SECTION 1

Linear SVM Classification

- **Large margin:** A linear SVM seeks a widest feasible street between classes, relying on points at the margin edge—the support vectors.
- **Soft margin:** Hyperparameter C trades margin width against slack violations, softening hard separability and reducing outlier sensitivity.
- **Scaling:** Feature scale mismatch skews margins; `StandardScaler` in a `Pipeline` is standard before `LinearSVC` or related linear trainers.
- **API variants:** `LinearSVC`, `SVC(kernel="linear")`, and hinge-loss `SGDClassifier` can all implement linear SVM-style classifiers with different speed profiles.

SECTION 2

Nonlinear SVM Classification

- **Feature expansion:** Polynomial features or similarity landmarks can make nonlinear boundaries in the original space linearly separable in a lifted space.
- **Kernel trick:** Kernelized `SVC` computes decisions as if features were mapped, using kernel functions (polynomial, RBF, etc.) without building huge explicit matrices.
- **RBF intuition:** Gaussian kernels measure closeness to training points; larger γ yields tighter, more complex boundaries; smaller γ smooths the decision surface.
- **Model choice:** Try linear first for scale, then RBF on smaller problems; grid search with cross-validation narrows *C*, *γ*, and degree settings.

SECTION 3

SVM Regression

- ε**-insensitive tube:** `LinearSVR` and `SVR` fit a band around predictions where errors inside width ε do not change the solution.
- **Regularization via C:** As in classification, C controls tolerance of points outside the tube, balancing fit tightness and generalization.
- **Kernelized regression:** `SVR` with polynomial or RBF kernels captures nonlinear trends analogous to kernel classifiers, at higher training cost on large *m*.

SECTION 4

Under the Hood

- **Decision function:** Classifiers score w■x + b; sign determines the predicted class without native probability outputs in basic SVM trainers.
- **Primal vs. dual:** Optimization can minimize weight norm subject to margin constraints or solve an equivalent dual problem in α coefficients over training pairs.
- **Kernelized predictions:** At inference, scores depend on kernels between the new point and support vectors, so only stored support patterns matter—not the full weight vector in feature space.
- **Complexity awareness:** Liblinear-style linear solvers scale roughly $O(m \times n)$; kernel SVC training often grows between $O(m^2 n)$ and $O(m^3 n)$, favoring smaller but rich datasets.